

Time Series Analysis

A. R. Amiri-Simkooei

Department of Geomatics Engineering

University of Isfahan



Parametric vs. non-parametric methods

Comparison

- **Parametric time series analysis**
 - Consists of trend, signal and noise
 - Functional effect modelling
 - Stochastic effect modelling
- **Non-parametric time series analysis**
 - Singular spectrum analysis (SSA)
 - Fourier transform
 - Wavelet transform
 - Machine learning methods



Singular spectrum analysis (SSA)

Introduction to SSA

- SSA applied to meteorological, ecological, economical and medical data sets.
- SSA decomposes original series into to some interpretable components.
- SSA decomposition is as: $y_i = s_i + n_i, i = 1, \dots, m$
- Components: slowly varying trend, oscillatory components and noise.
- SSA is a nonparametric spectral estimation method.
 - Neither a parametric model nor stationarity have to be assumed.
 - SSA is model-free and hence we expect a wide range of applications.
- Its name relates to the spectrum of eigenvalues in a SVD of a covariance matrix.
- There is no direct relationship to the frequency domain decomposition.
- It is based on SVD of a specific (trajectory) matrix.
- SSA and multichannel SSA (M-SSA) can handle nonlinear dynamics.
- SSA and M-SSA can handle spatio-temporal gap filling and prediction.
- SSA statistical testing is based on the Monte-Carlo SSA



Singular spectrum analysis (SSA)

Steps to implement SSA

- Decomposition
 1. Embedding
 2. Singular value decomposition (SVD)
- Reconstruction
 3. Eigentriple Grouping
 4. Diagonal averaging



Singular spectrum analysis (SSA)

Step 1: Embedding

Let $\mathbf{y} = [y_1, y_2, \dots, y_m]^T$ be a time series of length m . Given a window length L ($1 < L < m$), we may construct L -lagged vectors $\mathbf{y}_i = [y_i, y_{i+1}, \dots, y_{i+L-1}]^T$, $i = 1, 2, \dots, K$. The $L \times K$ trajectory matrix Y is defined as

$$Y = \begin{bmatrix} y_1 & y_2 & y_3 & \cdots & y_K \\ y_2 & y_3 & y_4 & \cdots & y_{K+1} \\ y_3 & y_4 & y_5 & \cdots & y_{K+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_L & y_{L+1} & y_{L+2} & \cdots & y_m \end{bmatrix} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_K]$$

where

$L = m - K + 1$ or $K = m - L + 1$ and Y is a Hankel matrix.

- In a Hankel matrix, the entries along a particular diagonal orthogonal to the main diagonal (or parallel to the main anti-diagonal) are equal.
- The entries of a Hankel matrix are $h_{ij} = y_{i+j-1}$, e.g., $h_{23} = y_{2+3-1} = y_4 = h_{32} = h_{14} = h_{41}$



Singular spectrum analysis (SSA)

Step 1: Embedding

- Embedding is a mapping from 1D vector to 2D $L \times K$ matrix
- Matrix Y^T is a $K \times L$ trajectory matrix with a window length equal to K
- The window length L of the matrix Y should be selected large enough.
 - To be a good representative of the entire time series
 - To be able to extract the required information
 - The length L expresses the dynamics of a system with the same length



Singular spectrum analysis (SSA)

Theoretical background

- The columns y_i (of Y) can be considered as vectors in L -dimensional space $y_i \in \mathbb{R}^L$. The eigenvalue decomposition of matrix YY^T (or equivalently, the SVD of matrix Y) yields a collection of L eigenvalues and eigenvectors.
- A particular combination of a certain number r of these eigenvectors determines an r -dimensional subspace $\mathbb{S}^r \subset \mathbb{R}^L$, where $r < L$. The L -dimensional data $\{y_1, y_2, \dots, y_K\}$ is then projected onto the subspace $\mathbb{S}^r$.
- The subsequent averaging over the diagonals yields some Hankel matrix $\tilde{Y}$. The time series $\tilde{y} = [\tilde{y}_1, \tilde{y}_2, \dots, \tilde{y}_m]^T$ which is in the one-to-one correspondence with matrix $\tilde{Y}$, provides an approximation either the whole series y or a particular component of that.



Singular spectrum analysis (SSA)

Step 2: Singular value decomposition (SVD)

- Set $S = YY^T$, a so-called $L \times L$ covariance matrix.
- Use eigenvalue decomposition of $S = U\Lambda U^T = \text{svd}(S)$.
 - Assume eigenvalues are in decreasing order as: $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_L \geq 0$.
 - Assume the $L \times L$ matrix U is denoted as $U = [\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_L]$ (eigenvectors).
- Assume $d = \text{rank}(Y) = \max(i, \text{where } \lambda_i > 0)$.
 - Usually, for real-life time series, we have $d = L^* = \min(L, K)$.
- Define new $K \times 1$ vectors $\mathbf{v}_i = Y^T \mathbf{u}_i / \sqrt{\lambda_i}$, $i = 1, \dots, d$.
 - It means $V = Y^T U \Lambda^{-\frac{1}{2}}$, where Y has a SVD decomposition $Y = U \Lambda^{\frac{1}{2}} V^T$.
 - The collection $(\sqrt{\lambda_i}, \mathbf{u}_i, \mathbf{v}_i)$ is called eigen triple.
- It is translated as $Y = Y_1 + \dots + Y_d$, where $Y_i = \sqrt{\lambda_i} \mathbf{u}_i \mathbf{v}_i^T$ is an $L \times K$ matrix.
- Matrices $Y_i, i = 1, \dots, d$, called elementary matrices, have rank of 1.
- Note that we can directly obtain $Y = U \Lambda^{\frac{1}{2}} V^T = \sum_{i=1}^d \sqrt{\lambda_i} \mathbf{u}_i \mathbf{v}_i^T$.



Singular spectrum analysis (SSA)

Step 2: Singular value decomposition (SVD)

- If we set $S = Y^T Y$, a so-called $K \times K$ covariance matrix is obtained, where its eigenvalue decomposition is $S = V \Lambda V^T = \text{svd}(S)$.
- We know $\sqrt{\lambda_i}$ are the singular values of Y , and $\mathbf{u}_i$ and $\mathbf{v}_i$ are the left and right singular vectors of Y .
- If eigenvalues λ_i are distinct, then the representation $Y_1 + \dots + Y_d$ is unique.
- From $Y = U \Lambda^{\frac{1}{2}} V^T = \sum_{i=1}^d \sqrt{\lambda_i} \mathbf{u}_i \mathbf{v}_i^T$, it follows $Y^T = \sum_{i=1}^d \sqrt{\lambda_i} \mathbf{v}_i \mathbf{u}_i^T$, which gives $Y^T Y = V \Lambda V^T$.
- One optimal property: Among all matrices $Y^{(r)}$ of rank $r < d$ the sum $\sum_{i=1}^r Y_i$ is the best approximation to trajectory matrix Y . This means $\|Y - Y^{(r)}\|_F \rightarrow \min$.
 - Frobenius norm: $\|Y\|_F^2 = \text{tr}(Y^T Y)$
- For trajectory matrix Y we have: $\|Y\|_F^2 = \text{tr}(Y^T Y) = \text{tr}(V \Lambda V^T) = \text{tr}(\Lambda) = \sum_{i=1}^d \lambda_i$
- For trajectory matrix $Y_i, i = 1, \dots, d$, we have: $\|Y_i\|_F^2 = \text{tr}(\lambda_i \mathbf{v}_i \mathbf{u}_i^T \mathbf{u}_i \mathbf{v}_i^T) = \lambda_i$
- Contribution of individual Y_i in the decomposition $Y = \sum_{i=1}^d Y_i$ is $\lambda_i / \|Y\|_F^2$



Singular spectrum analysis (SSA)

Step 2: Singular value decomposition (SVD)

- We then have $\|Y - Y^{(r)}\|_F = \sum_{i=r+1}^d \lambda_i$, which is minimum.
- We had $Y = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_K]$. We know $\mathcal{R}(Y) = \mathcal{R}([\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_K]) = \mathcal{X}^{(L)} \subset \mathbb{R}^L$, called trajectory space of time series $\mathbf{y}$.
- From $Y = U \Lambda^{\frac{1}{2}} V^T$, it is known that $\mathcal{R}(Y) = \mathcal{R}(U)$, indicating that $U = [\mathbf{u}_1, \dots, \mathbf{u}_d]$ is an orthonormal bases for $\mathcal{X}^{(L)}$. We then have $\dim \mathcal{X}^{(L)} = d$.
- $Y = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_K] = \sum_{i=1}^d \sqrt{\lambda_i} \mathbf{u}_i \mathbf{v}_i^T = \sum_{i=1}^d \mathbf{u}_i \mathbf{z}_i^T$, with $\mathbf{z}_i = \sqrt{\lambda_i} \mathbf{v}_i$. This gives $\mathbf{y}_j = \sum_{i=1}^d z_{ji} \mathbf{u}_i$.
- We assume $r < d$. If we have $S^r \subset \mathbb{R}^L$, the $S^r = \text{span}([\mathbf{u}_1, \dots, \mathbf{u}_r])$ approximate $\mathbf{y}_i$'s in best way. The min of $\sum_{i=1}^K \text{dist}^2(\mathbf{y}_i, S^r)$ is attained at $S^r = \text{span}([\mathbf{u}_1, \dots, \mathbf{u}_r])$.
- The characteristics of best approximation of lagged-vectors is: $\sum_{i=1}^r \lambda_i / \sum_{i=1}^d \lambda_i$.
- Another optimal feature relates to the properties of directions determined by the eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_d$. Direction of the eigenvector $\mathbf{u}_i$ the i^{th} principal direction.
- We had $Y = \sum_{i=1}^d Y_i = \sum_{i=1}^d \mathbf{u}_i \mathbf{z}_i^T$. The elementary matrix $Y_i = \mathbf{u}_i \mathbf{z}_i^T$ is built up from the projections of the lagged vectors onto i^{th} direction.



Singular spectrum analysis (SSA)

Step 3: Eigentriple grouping (1)

- Indices $\{1, \dots, d\}$, are partitioned into k groups $\{I_1, \dots, I_k\} \rightarrow$ Eigentriple grouping
- Grouping is aimed to see how separable are the additive component. Similar indices are within a group.
- Each group I consists of some of the indices $I = \{i_1, \dots, i_p\} \rightarrow Y_I = Y_{i_1} + \dots + Y_{i_p}$.
- It then follows: $Y = Y_1 + \dots + Y_d = Y_{I_1} + \dots + Y_{I_k}$.
- If $k = d$ we have $I_j = \{j\} \rightarrow$ Elementary grouping.



Singular spectrum analysis (SSA)

Step 3: Eigentriple grouping (2)

- Let us assume $k = 2 \rightarrow I_1 = I = \{i_1, \dots, i_r\}$ and $I_2 = \{1, \dots, d\} \setminus I$. This is motivated as usually I_1 refer to **signal** components but I_2 to **noise** components.
- Further assume the time series $\mathbf{y}$ is the sum of two series: $\mathbf{y} = \mathbf{y}^{(1)} + \mathbf{y}^{(2)}$,
 $y_i = y_i^{(1)} + y_i^{(2)}$, $i = 1, \dots, m$.
- The $L \times K$ trajectory matrices are $\mathbf{y}$, $\mathbf{y}^{(1)}$ and $\mathbf{y}^{(2)}$ are denoted as $Y, Y^{(1)}$ and $Y^{(2)}$.
- We can simply verify: $Y = Y^{(1)} + Y^{(2)} = Y_1 + \dots + Y_d$.
- We say that $\mathbf{y}^{(1)}$ and $\mathbf{y}^{(2)}$ are separable if there exist indices $I \subset \{1, \dots, d\}$ such that $Y^{(1)} = \sum_{i \in I} Y_i$ and $Y^{(2)} = \sum_{i \notin I} Y_i$.
- The contribution of $Y^{(1)}$ is then $\sum_{i \in I} \lambda_i / \sum_{i=1}^d \lambda_i$, and so is for $Y^{(2)}$.



Singular spectrum analysis (SSA)

Step 3: Eigentriple grouping (3)

- Another interpretation: If $Y_{I_1} = \sum_{i \in I_1} Y_i = Y^{(1)}$ is a Hankel matrix, they are separable.
- If matrices Y_{I_1} and Y_{I_2} are close to Hankel matrix, they are approximately separable.
- **Conclusion:** The goal of grouping is to find index sets $I_1, \dots, I_k$ such that matrices $Y_{I_1}, \dots, Y_{I_k}$ are close to a Hankel matrix.
- **Interpretation:** $U = [u_1, \dots, u_d]$ is an orthonormal bases for $\mathcal{X}^{(L)}$. By the above separation we have $\mathcal{X}^{(L)} = \mathcal{X}^{(L,1)} \oplus \mathcal{X}^{(L,2)}$, with $\mathcal{X}^{(L,1)} = \text{span}(u_i, i \in I_1)$ and $\mathcal{X}^{(L,2)} = \text{span}(u_i, i \in I_2)$.
- **Separability** of $\mathbf{y}^{(1)}$ and $\mathbf{y}^{(2)}$ means that the matrix Y_{I_1} , whose columns are the projections of the lagged vectors $\mathbf{y}_1, \dots, \mathbf{y}_K$ onto the $\mathcal{X}^{(L,1)}$, is exactly the trajectory matrix $Y^{(1)}$ of the time series $\mathbf{y}^{(1)}$.



Singular spectrum analysis (SSA)

Step 4: Diagonal averaging (1)

- For the time series $\mathbf{y} = [y_1, y_2, \dots, y_m]^T$, with m observations, we defined the following $L \times K$ trajectory matrix Y :

$$Y = \begin{bmatrix} y_1 & y_2 & y_3 & \cdots & y_K \\ y_2 & y_3 & y_4 & \cdots & y_{K+1} \\ y_3 & y_4 & y_5 & \cdots & y_{K+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ y_L & y_{L+1} & y_{L+2} & \cdots & y_m \end{bmatrix} = [\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_K]$$

where

$L = m - K + 1$ or $K = m - L + 1$ and Y is a Hankel matrix.

- The matrix Y_I , with $I \subset \{1, \dots, d\}$, made as the reconstructed part (signal of interest) does not have a Hankel structure.
- This indicates that the reconstructed time series y_I cannot be obtained from Y_I . A **Hankelization** process is then needed.



Singular spectrum analysis (SSA)

Step 4: Diagonal averaging (2)

- Assume we have an $L \times K$ matrix $X = x_{ij}$, $i = 1, \dots, L$ and $j = 1, \dots, K$, which is not Hankelized. How to make it to have a Hankel structure.

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1K} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2K} \\ x_{31} & x_{32} & x_{33} & \cdots & x_{3K} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{L1} & x_{L2} & x_{L3} & \cdots & x_{LK} \end{bmatrix}$$

To be converted to:

$$\tilde{X} = \begin{bmatrix} x_1 & x_2 & x_3 & \cdots & x_K \\ x_2 & x_3 & x_4 & \cdots & x_{K+1} \\ x_3 & x_4 & x_5 & \cdots & x_{K+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_L & x_{L+1} & x_{L+2} & \cdots & x_m \end{bmatrix}$$

- We set $L^* = \min(L, K)$
- We set $K^* = \max(L, K)$
- We set $m = L + K - 1$

- Assume $x_{ij}^* = x_{ij}$, if $L < K$
- Assume $x_{ij}^* = x_{ji}$, if $L \geq K$

$$X \rightarrow \tilde{X}$$



Singular spectrum analysis (SSA)

Step 4: Diagonal averaging (3)

- By making the **diagonal averaging** we transfer the matrix X into the series $x = [x_1, x_2, \dots, x_m]^T$ using the following formula:
- Note $H_{ij} = h_{i+j-1}$ means $i + j - 1 = k \rightarrow j = k - i + 1$

$$x_k = \begin{cases} \frac{1}{k} \sum_{i=1}^k x_{i,k-i+1}^* & \text{for } 1 \leq k < L^* \\ \frac{1}{L^*} \sum_{i=1}^k x_{i,k-i+1}^* & \text{for } L^* \leq k < K^* \\ \frac{1}{m-k+1} \sum_{i=k-K^*+1}^{m-K^*+1} x_{i,k-i+1}^* & \text{for } L^* \leq k < K^* \end{cases}$$

- For $k = 1$, we have $\frac{1}{k} \sum_{i=1}^k x_{i,k-i+1}^* = x_{1,1-1+1}^* = x_{1,1}^*$
- For $k = 2$, we have $\frac{1}{2} \sum_{i=1}^k x_{i,k-i+1}^* = \frac{1}{2} (x_{1,2-1+1}^* + x_{2,2-2+1}^*) = \frac{1}{2} (x_{1,2}^* + x_{2,1}^*)$
- For $k = 3$, we have $\frac{1}{3} \sum_{i=1}^k x_{i,k-i+1}^* = \frac{1}{3} (x_{1,3}^* + x_{2,2}^* + x_{3,1}^*)$



Singular spectrum analysis (SSA)

Step 4: Diagonal averaging (4)

- Diagonal averaging of previous slide can be applied to Y_{I_i} to produce a reconstructed $\tilde{\mathbf{y}}^{(i)} = [\tilde{y}_1^{(i)}, \dots, \tilde{y}_m^{(i)}]^T$.
- The initial time series $\mathbf{y} = [y_1, y_2, \dots, y_m]^T$ is therefore decomposed into a sum of k reconstructed series as $\mathbf{y} = \sum_{i=1}^k \tilde{\mathbf{y}}^{(i)}$.
- Henkelized matrix has the closest distance (among all Henkel matrices) to Y_{I_i} in the sense of Frobenius norm.
- There are techniques, called Henkelization operators, that can be used to make a Henkelized matrix.

